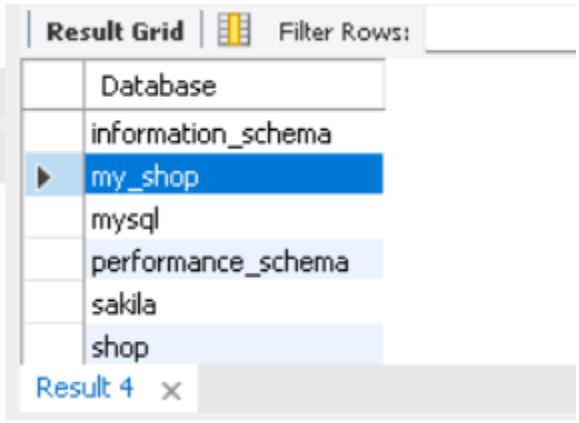
PRACTICE SET – 1

1. Create the shop database and the products table exactly as shown.

Confirm with SHOW TABLES;

```
1  CREATE DATABASE my_shop ;
2  SHOW DATABASES;
3  USE my_shop ;
4
```

DATABASE SCREENSHOT:



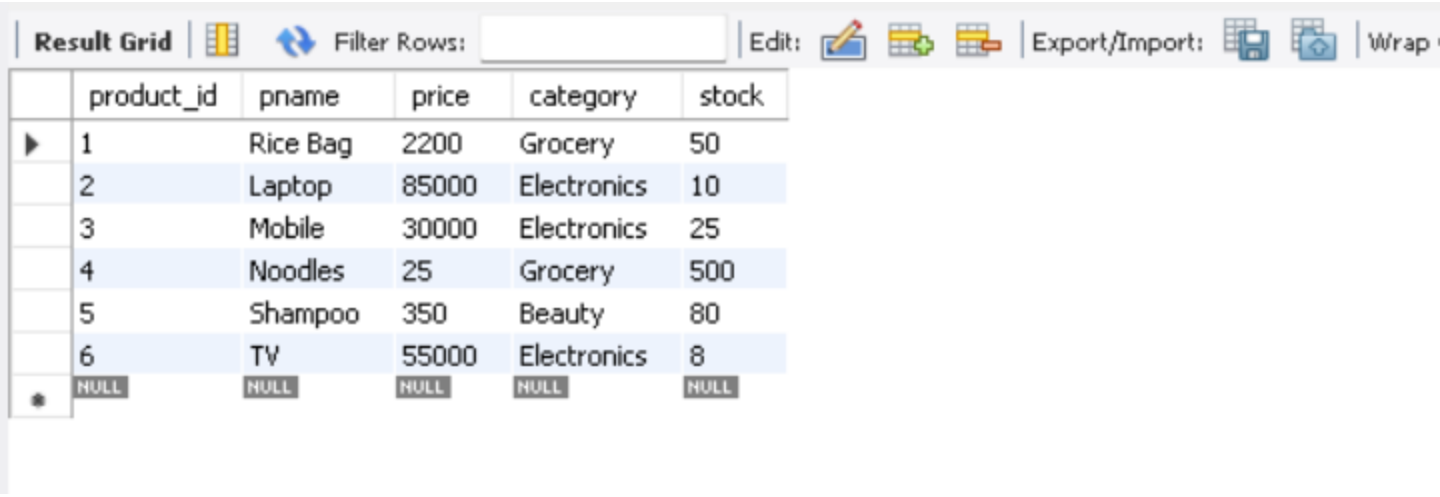
2. Insert all 6 products in one single INSERT statement.

```sql
CREATE TABLE products (
product_id INT PRIMARY KEY ,
pname VARCHAR (50) ,
price INT ,
category VARCHAR (30) ,
stock INT
);

SELECT * FROM products ;


INSERT INTO products
VALUES
(1,"Rice Bag",2200,"Grocery",50),
(2,"Laptop",85000,"Electronics",10),
(3,"Mobile",30000,"Electronics",25),
(4,"Noodles",25,"Grocery",500),
(5,"Shampoo",350,"Beauty",80),
(6,"TV",55000,"Electronics",8);
```

DATABASE OUTPUT:

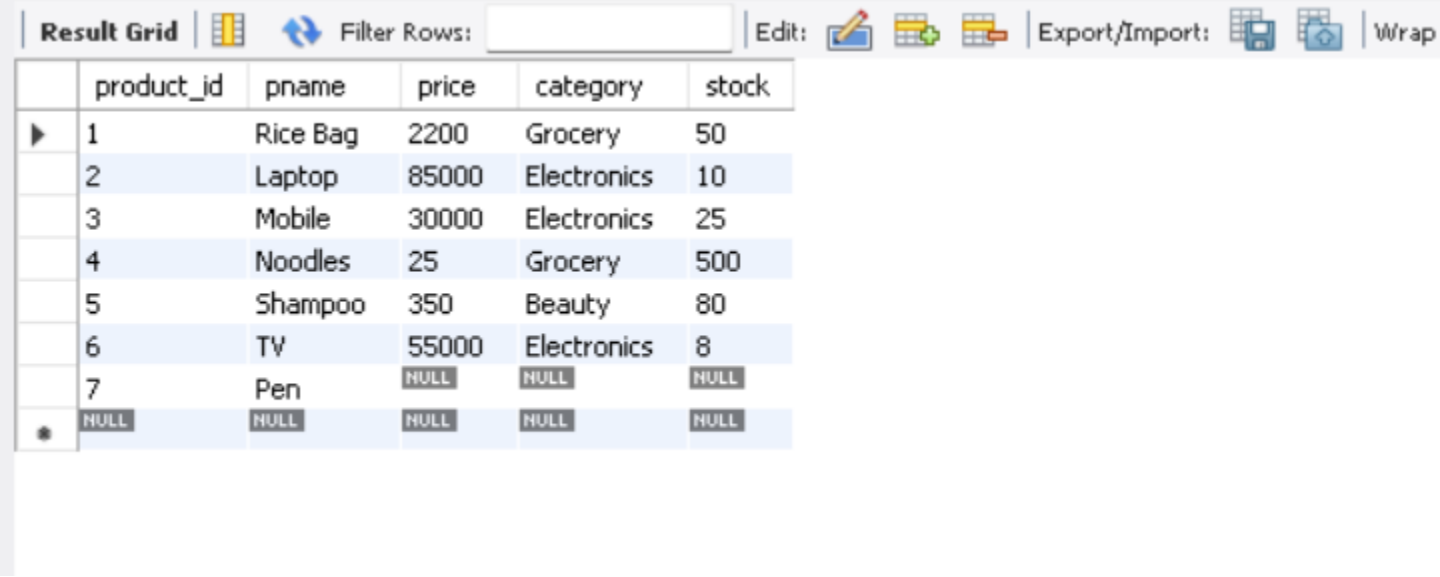


product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	50
2	Laptop	85000	Electronics	10
3	Mobile	30000	Electronics	25
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80
6	TV	55000	Electronics	8
NULL	NULL	NULL	NULL	NULL

3. Insert product 7, giving only product id and pname ('Pen'). Which columns become NULL?

```
28
29 •    INSERT INTO products(product_id,pname)
30      VALUES(7,"Pen");
```

DATABASE OUTPUT:



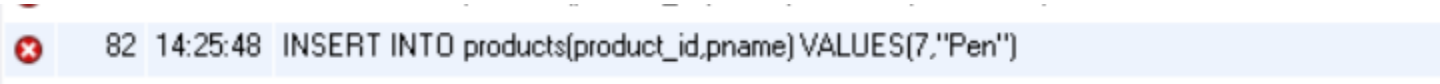
Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	50
2	Laptop	85000	Electronics	10
3	Mobile	30000	Electronics	25
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80
6	TV	55000	Electronics	8
7	Pen	NULL	NULL	NULL
NULL	NULL	NULL	NULL	NULL

4. Try to insert another product with product id = 3. Write down the exact error and explain why it happened.

```
31
32 •    INSERT INTO products(product_id,pname)
33      VALUES(3,"Mouse");
34
```

DATABASE OUTPUT:

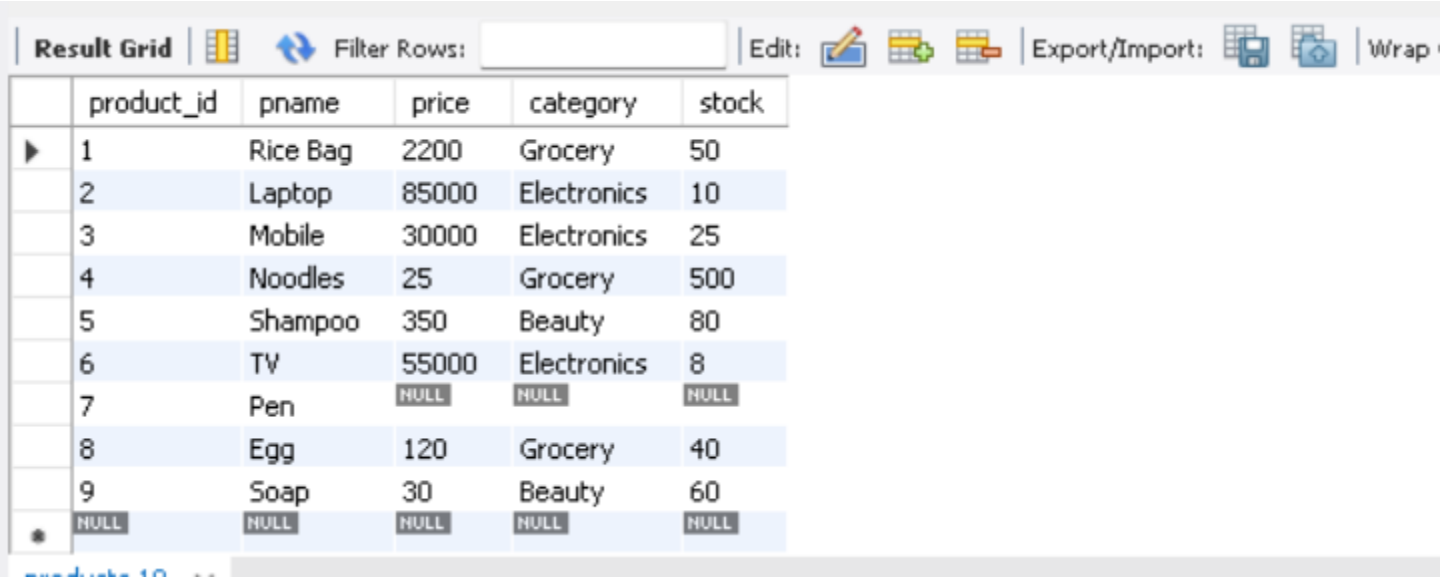


82 14:25:48 INSERT INTO products(product_id,pname) VALUES(7,"Pen")

5. Insert two more products of your choice in one statement: one Grocery item and one Beauty item.

```
34
35 •    INSERT INTO products
36      VALUES
37      (8,"Egg",120,"Grocery",40),
38      (9,"Soap",30,"Beauty",60);
39
```

DATABASE OUTPUT:



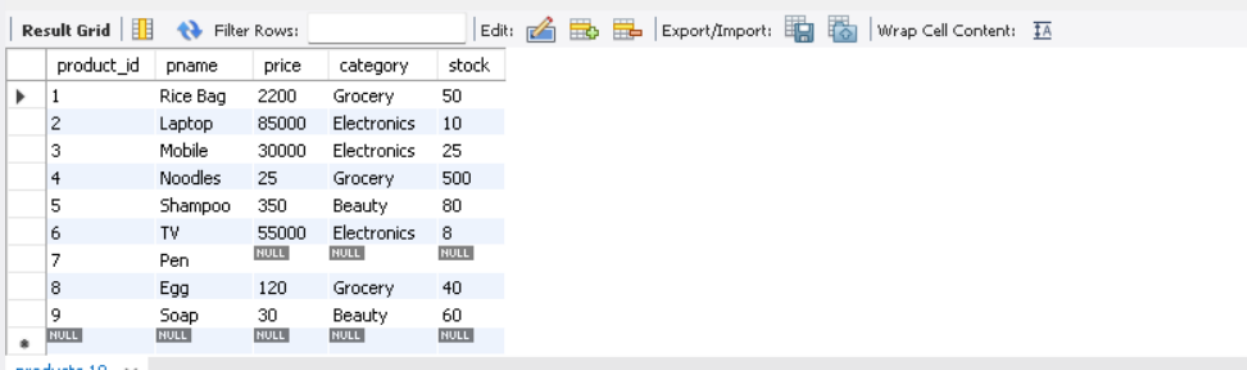
Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	50
2	Laptop	85000	Electronics	10
3	Mobile	30000	Electronics	25
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80
6	TV	55000	Electronics	8
7	Pen	NULL	NULL	NULL
8	Egg	120	Grocery	40
9	Soap	30	Beauty	60
NULL	NULL	NULL	NULL	NULL

6. Show all columns of all products.

```
40 •    SELECT * FROM products ;
41
```

DATABASE OUTPUT:



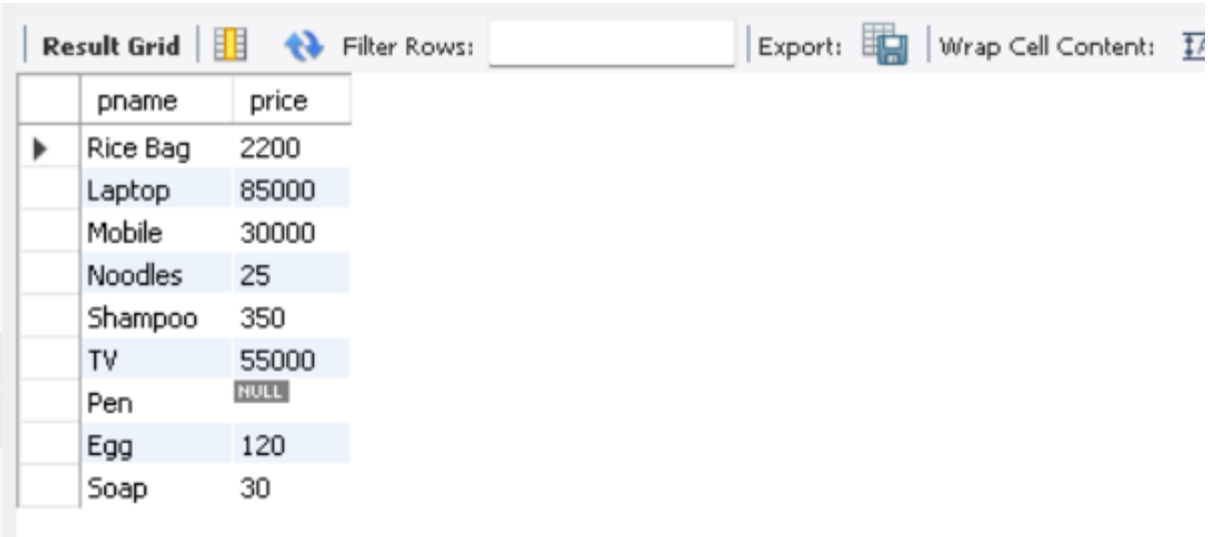
Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	50
2	Laptop	85000	Electronics	10
3	Mobile	30000	Electronics	25
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80
6	TV	55000	Electronics	8
7	Pen	NULL	NULL	NULL
8	Egg	120	Grocery	40
9	Soap	30	Beauty	60
NULL	NULL	NULL	NULL	NULL

7. Show only pname and price.

```
41
42 •    SELECT pname,price FROM products ;
43
```

DATABASE OUTPUT:



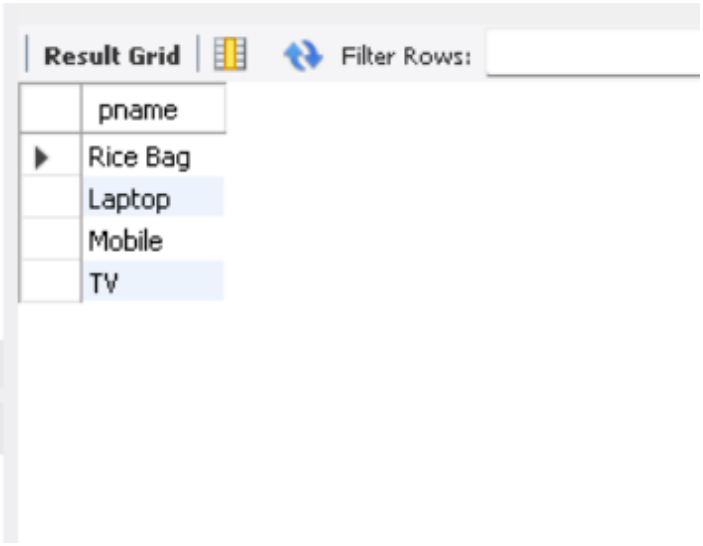
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

pname	price
Rice Bag	2200
Laptop	85000
Mobile	30000
Noodles	25
Shampoo	350
TV	55000
Pen	NULL
Egg	120
Soap	30

8. Show products with price above 1000.

```
43
44 •    SELECT pname FROM products WHERE price > 1000 ;
45
```

DATABASE OUTPUT:



9. Show Electronics products with stock less than 20 (AND).

```
45
46 •    SELECT pname FROM products WHERE category = "Electronics" AND stock < 20 ;
```

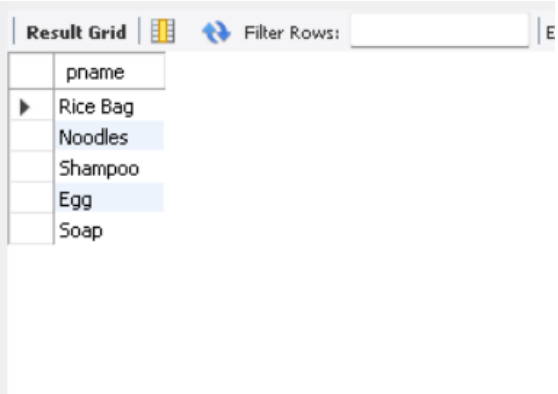
DATABASE OUTPUT:



10. Show products that are Grocery or cost less than 500 (OR).

```
47
48 •  SELECT pname FROM products WHERE category = "Grocery"  OR price < 500;
49
```

DATABASE OUTPUT:



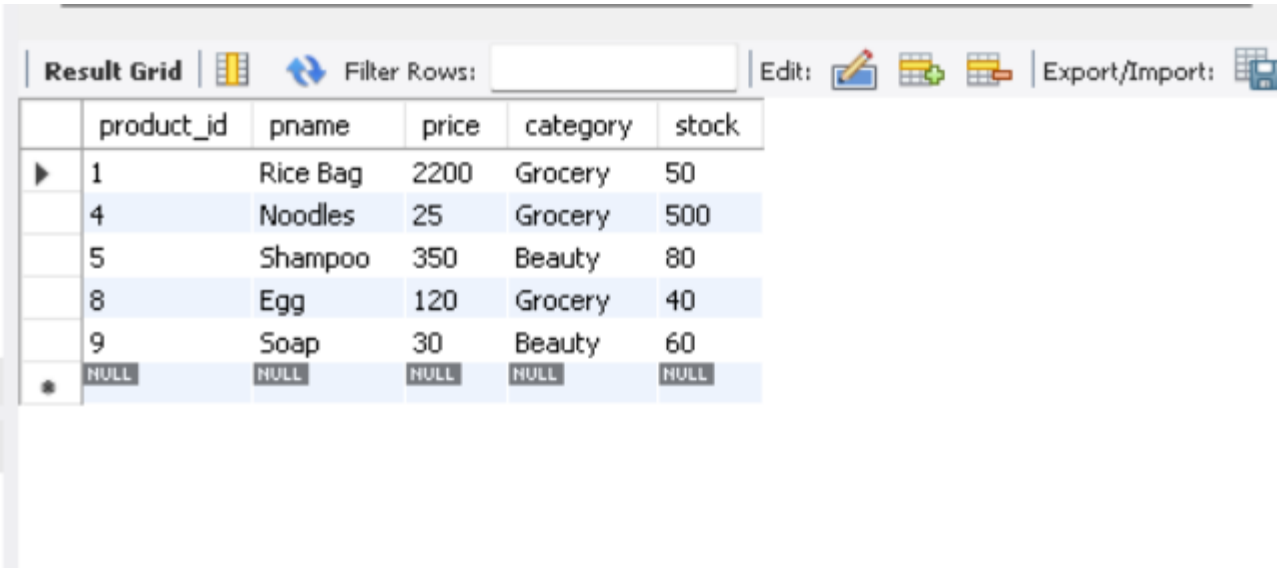
Result Grid | Filter Rows:

pname
Rice Bag
Noodles
Shampoo
Egg
Soap

11. Show products in categories Grocery and Beauty, using IN.

```
49
50 •  SELECT * FROM products WHERE category IN ("Grocery","Beauty");
51
```

DATABASE OUTPUT:



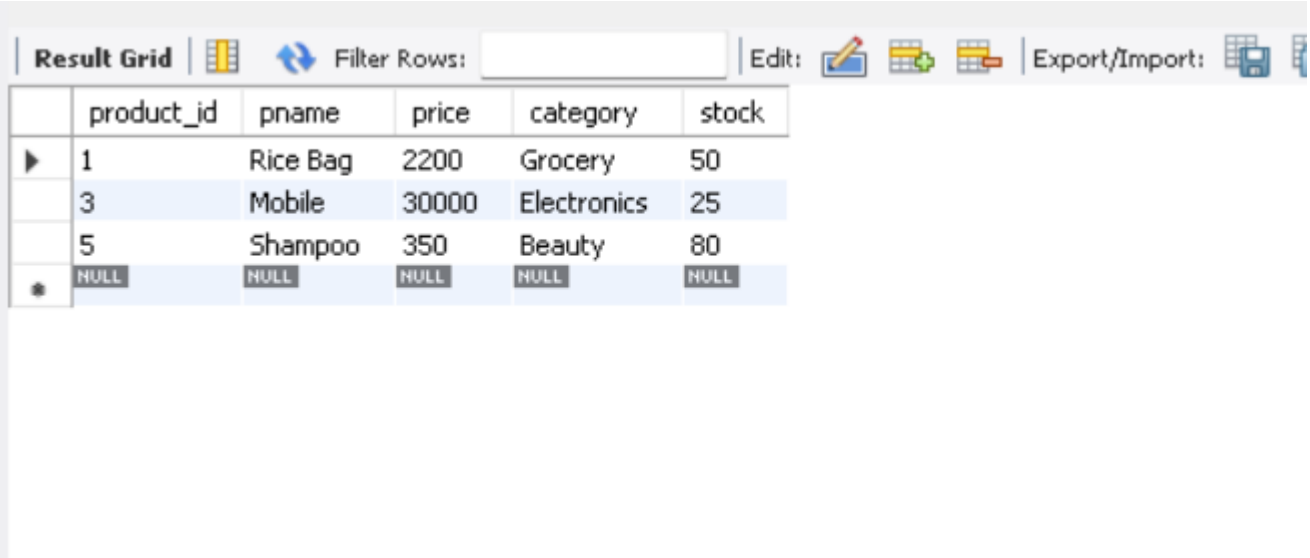
Result Grid | Filter Rows: | Edit: | Export/Import:

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	50
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80
8	Egg	120	Grocery	40
9	Soap	30	Beauty	60
NULL	NULL	NULL	NULL	NULL

12. Show products with price between 300 and 40000. Is 350 included? Why?

```
51
52 •  SELECT * FROM products WHERE PRICE BETWEEN 300 AND 40000 ;
53
```

DATABASE OUTPUT:



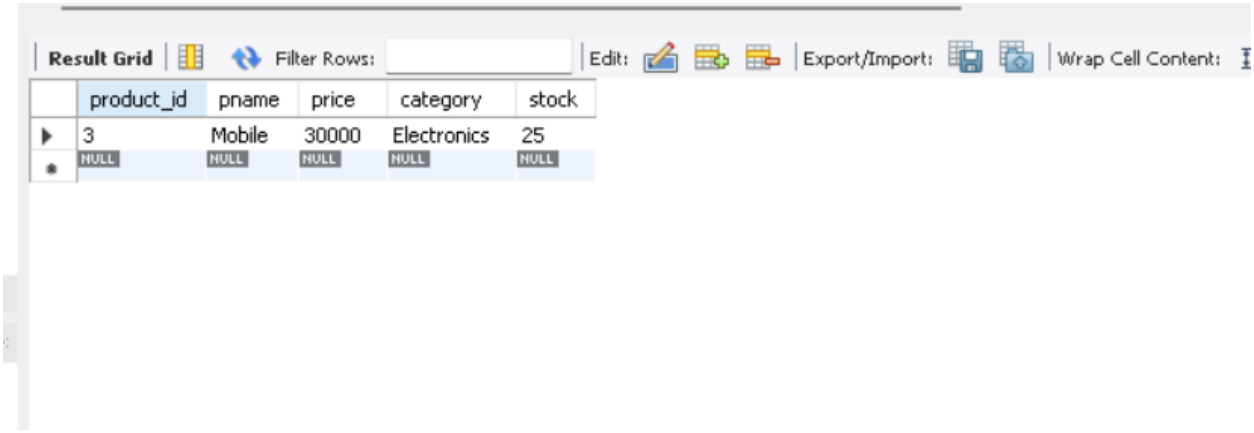
Result Grid | Filter Rows: | Edit: | Export/Import:

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	50
3	Mobile	30000	Electronics	25
5	Shampoo	350	Beauty	80
NULL	NULL	NULL	NULL	NULL

13. Show products whose name starts with M.

```
53
54 •  SELECT * FROM products WHeRE pname LIKE"M%" ;
```

DATABASE OUTPUT:



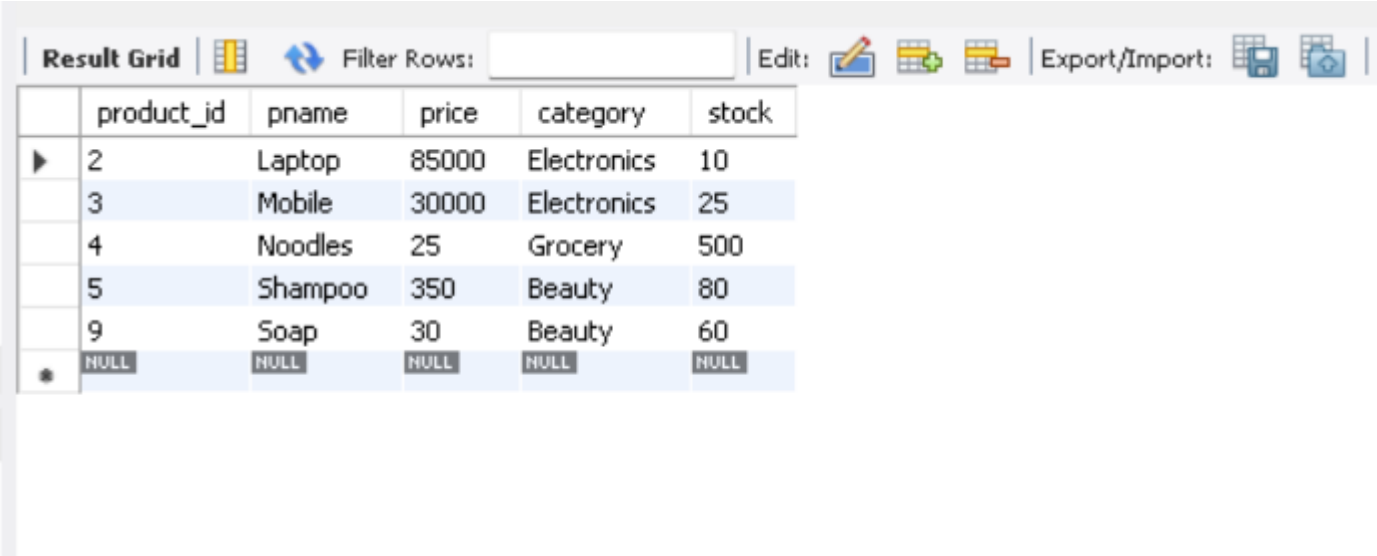
Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	pname	price	category	stock
3	Mobile	30000	Electronics	25
NULL	NULL	NULL	NULL	NULL

14. Show products whose name contains the letter o.

```
55
56 •    SELECT * FROM products WHERE pname LIKE "%o%";
```

DATABASE OUTPUT:



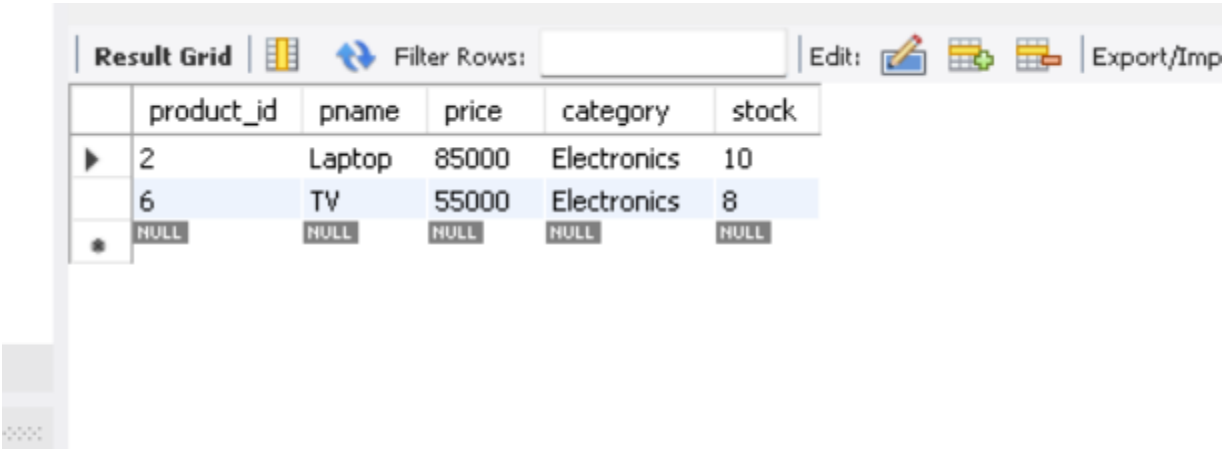
Result Grid | Filter Rows: | Edit: | Export/Import:

product_id	pname	price	category	stock
2	Laptop	85000	Electronics	10
3	Mobile	30000	Electronics	25
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80
9	Soap	30	Beauty	60
NULL	NULL	NULL	NULL	NULL

15. Show the 2 most expensive products (which two keywords together?).

```
57
58 •    SELECT * FROM products ORDER BY price DESC LIMIT 2 ;
```

DATABASE OUTPUT:



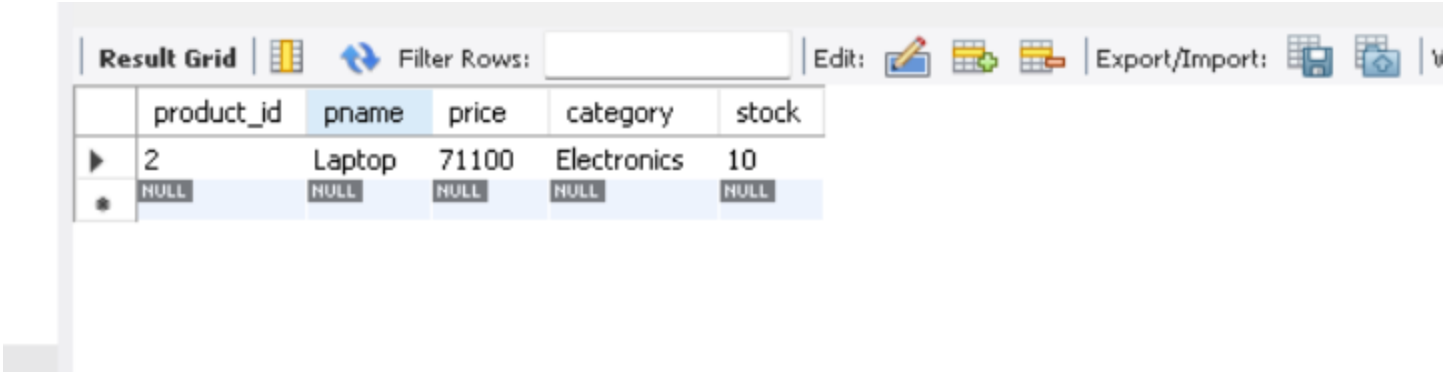
Result Grid | Filter Rows: | Edit: | Export/Imp

product_id	pname	price	category	stock
2	Laptop	85000	Electronics	10
6	TV	55000	Electronics	8
NULL	NULL	NULL	NULL	NULL

16. The Laptop price dropped to 79000. Update it using its id.

```
59
60 •  UPDATE products
61    SET price = 79000
62    WHERE product_id = 2;
63
```

DATABASE OUTPUT:



Result Grid | Filter Rows: | Edit: | Export/Import:

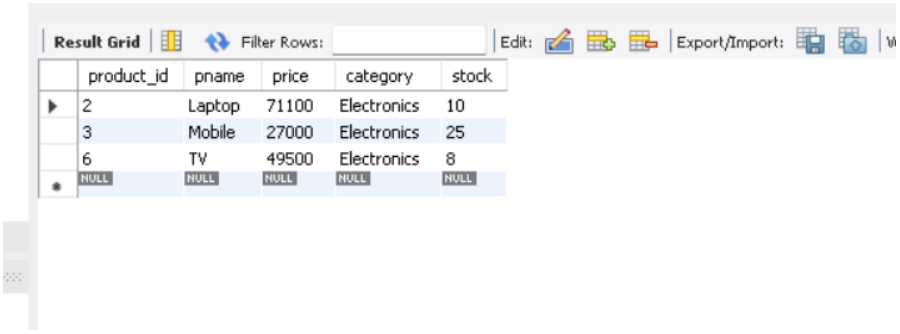
product_id	pname	price	category	stock
2	Laptop	71100	Electronics	10
NULL	NULL	NULL	NULL	NULL

17. Festival offer: reduce the price of every Electronics product by 10%.

Hint: SET price = price * 0.9.

```
65
66 •  UPDATE products
67    SET price = price * 0.9
68    WHERE product_id IN (2,3,6) ;
69
```

DATABASE OUTPUT:



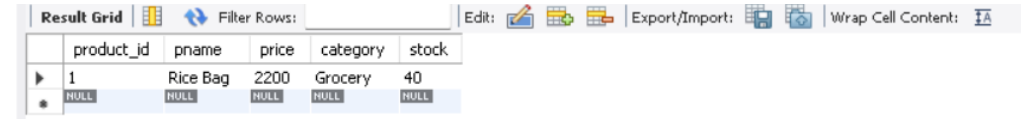
Result Grid | Filter Rows: | Edit: | Export/Import:

product_id	pname	price	category	stock
2	Laptop	71100	Electronics	10
3	Mobile	27000	Electronics	25
6	TV	49500	Electronics	8
NULL	NULL	NULL	NULL	NULL

18. Ten Rice Bags were sold. Decrease its stock by 10 with one query.

```
71
72 •    UPDATE products
73      SET stock = stock - 10
74      WHERE pname = "Rice Bag" ;
75
```

DATABASE OUTPUT:



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	40
NULL	NULL	NULL	NULL	NULL

19. Delete the product named Pen. Then delete all products with stock below

10. How many rows did each query remove?

```
77
78 •    DELETE FROM products WHERE pname = 'Pen' ;
79
80 •    DELETE FROM products WHERE stock < 10 ;
81
```

DATABASE OUTPUT:

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	40
2	Laptop	71100	Electronics	10
3	Mobile	27000	Electronics	25
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80
8	Egg	120	Grocery	40
9	Soap	30	Beauty	60
NULL	NULL	NULL	NULL	NULL

20. Run SELECT to check the final table. Then answer: which command would empty the table but keep it, and which would remove the table completely?

```
82 •    SELECT * FROM products ;
```

DATABASE OUTPUT:

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	40
2	Laptop	71100	Electronics	10
3	Mobile	27000	Electronics	25
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80
8	Egg	120	Grocery	40
9	Soap	30	Beauty	60
NULL	NULL	NULL	NULL	NULL